

## Project Sprint #1

The Solitaire game is described in CS449HomeworkOverview. You should read the description very carefully.

In this assignment, you aim to specify the requirements (i.e., user stories and acceptance criteria) of the target software that allows a human player to play a Solitaire game. These requirements will be fully implemented by the end of sprint 3. The minimum features include **choosing the board size and type, starting a new game, making a move, determining if a game is over**. The following is a sample GUI layout.

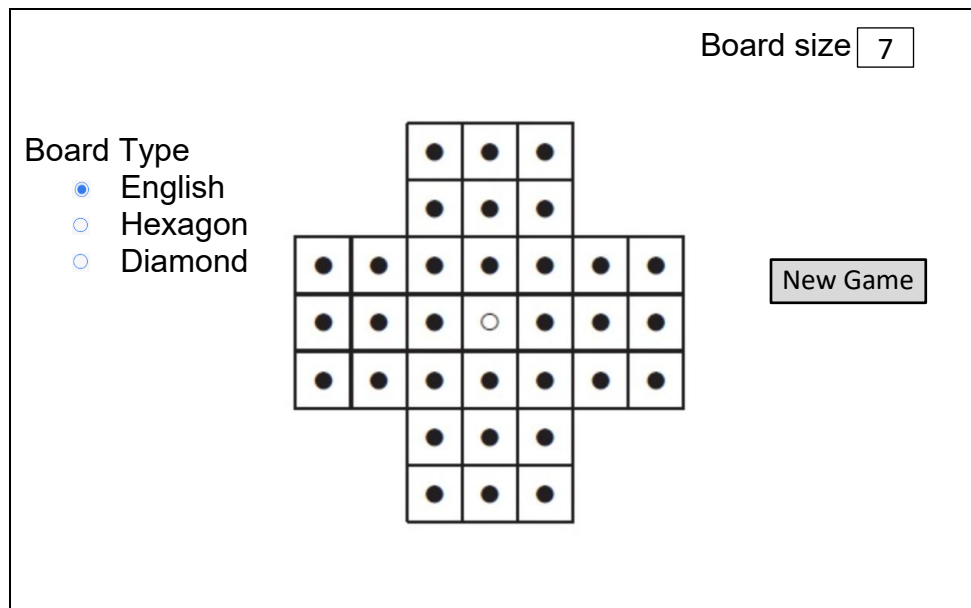


Figure 1. Sample GUI layout of the first working program by the end of Sprint 3

Use the following tables to document your user stories and acceptance criteria.

You are required to use the free ChatGPT or another LLM to complete two user stories and two of their respective acceptance criteria. You must ensure that the generated user stories are correct and fix them if not. At the end of the submission, provide screenshots of your LLM prompts and answers, along with the description of errors the LLM made that you had to correct. Points will be deducted if no screenshots/description are provided.

For the other user stories and acceptance criteria, you are not required to use LLM.

### I. User Stories (3 points)

- **User Story Template:** As a <role>, I want <goal> [so that <benefit>]

| ID | User Story Name       | User Story Description                                                                                                                   | Priority | Estimated effort (hours) |
|----|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------|
| 1  | Choose a board size   | As a player, I want to choose the size of the Peg Solitaire board, so that I can customize the game to my preferred level of complexity. | High     | 2                        |
| 2  | Choose the board type | As a player, I want to choose the type of Peg Solitaire board, so that I can play using my preferred board layout.                       | High     | 3                        |

| ID | User Story Name                                    | User Story Description                                                                                                                              | Priority | Estimated effort (hours) |
|----|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------|
| 3  | Start a new game of the chosen board size and type | As a player, I want to start a new game using my selected board size and board type, so that I can play Peg Solitaire with my chosen configuration. | High     | 4                        |
| 4  | Make a move in a game                              | As a player, I want to make valid moves on the Peg Solitaire board, so that I can remove pegs and progress through the game.                        | High     | 5                        |
| 5  | A game is over                                     | As a player, I want the system to detect when the game is over, so that I know when I have finished playing and how well I performed.               | High     | 3                        |

## II. Acceptance Criteria (AC) (12 points): Add/delete rows as needed.

| User Story ID and Name                                | AC ID | Description of Acceptance Criterion                                                                                                                                                                                                                                                                                | Status (completed, toDo, inProgress) |
|-------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 1. Choose a board size                                | 1.1   | AC 1.1 <Display available board sizes><br>Given the player is setting up a new game,<br>When the player views the board size options,<br>Then the system displays all available board sizes that can be selected.                                                                                                  |                                      |
|                                                       | 1.2   | AC 1.2 <Select a board size><br>Given the available board sizes are displayed,<br>When the player selects a board size,<br>Then the system records the selected size for the new game.                                                                                                                             |                                      |
|                                                       |       |                                                                                                                                                                                                                                                                                                                    |                                      |
| 2. Choose the board type                              | 2.1   | AC 2.1 <Display Board type options><br>Given the player is setting up a new game,<br>When the player views the board type options,<br>Then the system displays English, Hexagon, and Diamond as the available board types.                                                                                         |                                      |
|                                                       | 2.2   | A.C 2.2 <Select board type><br>Given the available board types are displayed,<br>When the player selects English, Hexagon, or Diamond,<br>Then the system records the selected board type for the new game.                                                                                                        |                                      |
| 3. Start a new game of the chosen board size and type | 3.1   | A.C 3.1 <Start game with selected config><br>Given the player has selected a valid board size and board type,<br>When the player selects New Game,<br>Then the system creates a new Peg Solitaire board using the selected board size and board type.                                                              |                                      |
|                                                       | 3.2   | A.C 3.2 <Initialize the board><br>Given the player has selected a valid board size and board type,<br>When a new game is started,<br>Then the system initializes the board with pegs in all playable positions except the center position, which is empty.                                                         |                                      |
|                                                       | 3.3   | A.C 3.3 <Prevent starting without req selections><br>Given the player has not selected a valid board size or board type,<br>When the player attempts to start a new game,<br>Then the system does not start the game and prompts the player to make the required selection.                                        |                                      |
| 4. Make a move                                        | 4.1   | AC 4.1 <Make a valid move><br>Given a game is in progress and a peg has an adjacent peg with an empty hole two positions away,<br>When the player moves the peg over the adjacent peg into the empty hole,<br>Then the system moves the selected peg into the empty hole and removes the peg that was jumped over. |                                      |

| User Story ID and Name | AC ID | Description of Acceptance Criterion                                                                                                                                                                                                                                                          | Status (completed, toDo, inProgress) |
|------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
|                        | 4.2   | A.C 4.2 <Allow orthogonal or diagonal moves><br>Given a game is in progress and a valid move is available,<br>When the player makes a move orthogonally or diagonally over an adjacent peg into an empty hole two positions away,<br>Then the system accepts the move and updates the board. |                                      |
|                        | 4.3   | A.C 4.2 <reject invalid move><br>Given a game is in progress,<br>When the player attempts a move that does not jump over an adjacent peg into an empty hole two positions away,<br>Then the system rejects the move and leaves the board unchanged.                                          |                                      |
| 5. A game is over      | 5.1   | A.C 5.1 <Game ends with one peg remaining><br>Given a game is in progress,<br>When only one peg remains on the board,<br>Then the system determines that the game is over.                                                                                                                   |                                      |
|                        | 5.2   | A.C 5.2 <Game ends with no valid moves remaining><br>Given a game is in progress with more than one peg remaining,<br>When no valid moves are available,<br>Then the system automatically determines that the game is over.                                                                  |                                      |
|                        | 5.3   | A.C 5.3 <Display players score><br>Given the system has determined that the game is over,<br>When the system counts the number of pegs remaining,<br>Then the system displays the player's rating based on the number of remaining pegs.                                                     |                                      |
|                        |       |                                                                                                                                                                                                                                                                                              |                                      |

The generated response put "generate the selected board" and "prevent starting the game" under the board-selection stories, but I recognized that those behaviors fit better under User Story 3: Start a new game of the chosen board size and type. Please see below.

I am working on a software development project for a Peg Solitaire game. I need to write user stories and acceptance criteria for two features: "Choose a board size" and "Choose the board type." Please create one user story for each feature using the format: "As a <role>, I want <goal>, so that <benefit>" Then create appropriate acceptance criteria for each user story using the given-when-then format. The available board types are English, Hexagon, and Diamond. Please make the acceptance criteria specific and testable

Absolutely. Since these are for a software development project, the acceptance criteria should describe **observable behavior that your team can actually test**. I'd write them like this:

## User Story 1 — Choose a Board Size

### User Story:

As a player, I want to choose the size of the Peg Solitaire board, so that I can customize the game to my preferred level of complexity.

### Acceptance Criteria:

#### AC1 — Display available board sizes

- **Given** the player is setting up a new game,
- **When** the player views the board size options,
- **Then** the system displays all available board sizes that can be selected.

#### AC2 — Select a board size

- **Given** the available board sizes are displayed,
- **When** the player selects a board size,
- **Then** the system records the selected size for the new game.

#### AC3 — Generate the selected board size

- **Given** the player has selected a valid board size, 
- **When** the player starts the game,

#### AC4 — Prevent starting without a valid size

- **Given** the game requires the player to select a board size,
  - **When** the player attempts to start the game without a valid board size selected,
  - **Then** the system prevents the game from starting and prompts the player to select a board size.
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## User Story 2 — Choose the Board Type

### User Story:

As a player, I want to choose the type of Peg Solitaire board, so that I can play using my preferred board layout.

### Acceptance Criteria:

#### AC1 — Display board type options

- **Given** the player is setting up a new game,
- **When** the player views the board type options,
- **Then** the system displays **English, Hexagon, and Diamond** as the available board types.

#### AC2 — Select a board type

- **Given** the available board types are displayed,
- **When** the player selects English, Hexagon, or Diamond,
- **Then** the system records the selected board type for the new game.

#### AC3 — Generate the selected board type

- **Given** the player has selected a board type,
- **When** the player starts the game,
- **Then** the system generates a Peg Solitaire board that matches the selected board type.

#### AC4 — Correct layout is displayed

- **Given** the player has selected English, Hexagon, or Diamond,
- **When** the game board is generated,
- **Then** the displayed board layout matches the ru  and structure of the selected board type.

#### AC5 — Prevent starting without a valid board type